National AI Hackathon 2026 by SchubKI

16-21 of October 2026 (16 – opening, 17-18 – work, 19-20 – evaluation, 21 - closing)

Art Fiber Systems

Challenge name

AI Assistant for In-line Spectroscopy Solution Design

Challenge description

Industrial customers often know what they need to monitor — for example concentration, reaction progress, contamination, moisture, or product quality — but they may not know which spectroscopy method, spectral range, fiber probe, or measurement setup should be tested first.

The goal is to build an AI assistant that helps an AFS expert make a first technical assessment of a new customer problem. The assistant should read a short use case, identify missing information, recommend one or more promising spectroscopy approaches, explain the recommendation, and propose a simple first feasibility experiment.

The assistant is not expected to replace an expert. It should help structure the first customer discussion and make recommendations that are transparent and easy to review.

Challenge (input + details)

Participants will receive a zip-archive (materials.zip):

- AFS presentations and short technical materials about spectroscopy methods, spectral ranges, fiber probes, and application examples:
    - S01_Multispectral_Laboratory.pdf
    - S02_AFS_Process_Control_Slides.pdf
    - S03_LED_Kidney_Study_2017.pdf
    - S04_Combi_MIR_Fluorescence_2021.pdf
    - S05_AFS_Flyer.pdf
    - S06_Laboratory_Link.txt (https://mirphotonics.com/laboratory/)
- A simple AFS reference table: method → suitable applications → advantages → limitations → relevant probe types:
    - S01_Multispectral_Laboratory.pdf (page 6)
    - AFS_method_comparison_table.xlsx
- **8 examples of customer scenarios for development:** AFS_Examples_and_Reference.docx (D01-D08)
- **6 customer scenarios for final evaluation:** customer_scenarios_for_final_evaluation.txt (H01 – H06).

For each customer scenario, the AI assistant should:

1. Extract the main process requirements and constraints.
2. Ask important clarification questions when information is missing.
3. Recommend one or more spectroscopy methods or combinations worth testing.
4. Suggest a high-level probe / measurement concept.
5. Explain the recommendation, limitations, and uncertainty.

6. Propose a short first feasibility-study plan.

Example

Example customer input	Expected assistant output
"We need to monitor moisture and hydrocarbon contamination in soil. Measurements should be remote and the environment may be humid and dirty."	Clarification questions; recommended method(s); relevant spectral range; possible probe concept; main risks; and a short plan for the first comparison experiment.

Minimal requirements

- Accept a customer/process description in natural language.
- Use the supplied AFS knowledge base as the main source for technical recommendations.
- Recommend at least one spectroscopy approach and explain why.
- Ask clarification questions when key information is missing.
- Provide a simple feasibility-study plan.
- Show the source or AFS material used for important recommendations.
- Provide a working prototype that can be demonstrated.
- **Rules.** External sources are allowed, but the supplied AFS materials remain the main source; an external citation must include the quoted passage. Any model or tool may be used. The answers to H01–H06 must be generated by your assistant (no manual editing, no case-specific hard-coding) and included in your repository. The jury may ask you to run a scenario or a short new enquiry live.

Expected output

- A working prototype: lightweight web app, notebook interface, or chat assistant.
- Git repository – Source code and a reproducible setup.
- Structured output for each case: requirements, questions, recommendation, risks, and feasibility plan.
- Short description of the system architecture and use of AFS materials.
- Final presentation and live demo (up to 5 mins) + slides.

Evaluation criteria

Criterion	What judges look for	Weight
Technical quality	Recommendations are reasonable for the customer case.	30%
Use of AFS knowledge	Recommendations are grounded in supplied AFS materials.	20%
Feasibility-study plan	The proposed first experiments are practical and useful.	15%
Clarification & uncertainty	The assistant asks good questions and knows when information is insufficient.	15%
Explainability	Reasoning is clear, traceable, and easy for an expert to review.	10%
Prototype & presentation	The solution works and is easy to demonstrate.	10%

Dataset / knowledge-base links

- **Dataset download:** the link